

Deniz Efe Karacakaya

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TECHNICAL SKILLS

AI & Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, SHAP (Explainable AI), Optuna (hyperparameter optimization), Federated Learning (Flower - FedAvg, FedProx, FedAdam, FedYogi, SCAFFOLD; Ditto, FedPer), Pandas, NumPy

ML Engineering & MLOps: MLflow (experiment tracking, model registry, alias-based model promotion), DVC (data versioning), Great Expectations (data validation), Evidently (data drift detection), KServe (Open Inference Protocol), model serving, automated retraining, reproducible training pipelines

Languages: Python, Java, TypeScript, JavaScript, SQL

Frameworks & APIs: FastAPI, Spring Boot, Flask, Node.js, Express, React, Next.js, Angular, Tailwind CSS, Hibernate/JPA, REST, gRPC, JWT

Cloud, Infrastructure & DevOps: Kubernetes, Docker, Docker Compose, Helm, Kustomize, Argo CD (GitOps), GitHub Actions (CI/CD), Actions Runner Controller, GHCR, Traefik, Google Cloud, Git, Linux

Databases & Storage: PostgreSQL, MinIO (S3-compatible object storage), MongoDB, Supabase, Firebase

Observability & Security: Prometheus, Grafana, Alertmanager, Sealed Secrets, Trivy (image vulnerability scanning), secret scanning in CI

EXPERIENCE

ASEE Information Technologies

AI Engineering Intern

Istanbul, Türkiye

Jun 2026 – Present

- Delivered **three consecutive proof-of-concept studies** for the **Banking R&D** fraud detection program — feature reduction, federated learning, and the MLOps platform beneath both — all shaped by one constraint: customer transaction data cannot leave the institution that holds it.
- Established the centralized reference on the **IEEE-CIS** dataset (~590K transactions, 391 features, severe class imbalance), benchmarking **Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost and an MLP** under a leakage-screened out-of-time protocol; **XGBoost** set the reference **F1 at 0.494**.
- Applied **SHAP** attribution to rank features and ran an iterative degradation test, reducing the schema **from 391 to 20 features while retaining 98.2% of reference F1** (0.485) and locating the cliff below ten features — shortening every retraining cycle and shrinking the feature pipeline that has to be produced, validated and monitored.
- Implemented the **federated learning** pipeline with **Flower and PyTorch over gRPC** on a **non-IID two-bank split** (partitioned by product line, 2.1% vs 7.2% fraud rate), and benchmarked **47 configurations** across FedAvg, FedProx, FedAdam, FedYogi and SCAFFOLD plus Ditto and FedPer personalization.
- Identified **fraud-prevalence weighting** as the deciding factor: the shared model **matched the strongest single bank** (0.3417 vs 0.3424 F1) while **lifting the weakest participant from 0.1667 to 0.3417 (+0.175 F1)** with no raw records exchanged, and reported the negative results that made the case (row-count weighting collapsed to 0.2507, SCAFFOLD failed at 0.1871).
- Engineered the **self-hosted Kubernetes MLOps platform** the studies run on, so that every experiment is traceable to the code, parameters and dataset version behind it: **MLflow** tracking and model registry on a **PostgreSQL** backend with **MinIO** artifact storage, **DVC** dataset versioning validated by **Great Expectations**, in-cluster CPU training jobs, and **Optuna** hyperparameter search logged as nested runs behind an **alias-based champion/challenger promotion gate**.
- Orchestrated the whole cluster through **GitOps** — **24 Argo CD Applications** in an app-of-apps hierarchy ordered by sync waves with prune and self-heal — and automated a **nine-job GitHub Actions pipeline** on in-cluster **ARC** runners covering linting, manifest rendering, Python tests, **Trivy** image scanning and two independent gates keeping plaintext credentials out of Git, with image tags committed back so no deployment is run by hand.
- Closed the loop from model to feedback: two serving paths (**FastAPI** and a **KServe InferenceService** over the **Open Inference Protocol**), **Prometheus/Grafana/Alertmanager** observability, and an **Evidently** drift job whose verdict raises an alert that creates a retraining Job — **verified end to end** against a deliberately perturbed dataset.
- Consolidated the three PoCs and the platform into a formal **corporate R&D project proposal** — technical scope, phase plan, KPIs and risk register — revised through supervising-engineer review, and documented the work as an internal technical reference.

PROJECTS

local-flow-mlops — Self-Hosted MLOps Platform

Built during the ASEE internship · github.com/denizefekaracakaya/local-flow-mlops

- Self-hosted, end-to-end machine learning platform on single-node **Kubernetes**, built for environments where data cannot leave the organisation — no managed cloud service anywhere in the workflow.
- **Cluster & delivery**: whole stack defined in Git and reconciled by **Argo CD** — app-of-apps with **24 Applications**, sync-wave ordering, prune and self-heal; **Helm** and **Kustomize** manifests, **Sealed Secrets** for every credential.
- **Data & training**: **DVC** dataset versioning against **MinIO**, **Great Expectations** validation in the ingestion path, in-cluster CPU training CronJob, and **Optuna** hyperparameter search as nested MLflow runs scored by 5-fold stratified cross-validation.
- **Tracking & registry**: **MLflow** on a **PostgreSQL** backend for run metadata and **MinIO** for artifacts, with **alias-based champion/challenger promotion** gated on a fixed improvement margin.
- **Serving**: **FastAPI** service resolving the champion alias per request, plus a **KServe** InferenceService on the **MLServer** runtime over the **Open Inference Protocol**.
- **Monitoring & automated retraining**: **Prometheus**, **Grafana** and **Alertmanager**; an **Evidently** drift job whose verdict raises an alert that creates a retraining Job through a token-authenticated in-cluster receiver.
- **CI/CD**: **nine-job GitHub Actions** pipeline on in-cluster **ARC** runners — YAML and shell linting, Helm/Kustomize rendering, Python tests, **Trivy** image scanning and two independent gates blocking plaintext secrets; image tags committed back to Git for Argo CD to roll out.
- **Cluster-free path**: **Docker Compose** bring-up of the training core, with training, tuning and drift checks as one-shot commands.

L.I.O.N.E.L — Local-First Voice Agent

Personal project · github.com/denizefekaracakaya/L.I.O.N.E.L

- Bilingual (English/Turkish), **MCP-native voice agent designed to run with the network unplugged**: four degradation tiers from fully offline to cloud-assisted, split across a low-latency host runtime and a relocatable cluster runtime, with **MCP** carrying control signals, **gRPC** the media streams and **OpenTelemetry** tracing.
- **Contracts before code**: 28 architecture decision records, 16 policy gates and 88 validation rules enforced in CI ahead of implementation — including a blocking conformance check for the fully offline tier and a hard rule against arbitrary shell execution anywhere in the system.

Chronos — Luxury Watch E-Commerce Platform

Academic full-stack project

- Architected a **role-based e-commerce platform** on Google Cloud — **Angular** front end, **Spring Boot** API, **JWT** authentication, **Stripe** payments and a Flask service exposing a **Gemini** chatbot hardened against prompt injection — over a normalized **(3NF)** schema loaded by a **Python ETL pipeline** (1,500+ product records).

Gym Buddy AI — Voice-Driven Fitness Coach

- Built an **AI fitness and nutrition assistant** delivering real-time voice guidance during workouts, on a **React/TypeScript** front end over a **Supabase** backend.

Artifree — Generative AI Photo Editor

- Developed a **generative photo-editing** web application supporting object removal and image enhancement on the **Gemini** API.

EDUCATION

Akdeniz University

B.S. in Artificial Intelligence & Data Engineering

Antalya, Türkiye

2024 – 2028 (Expected)

CERTIFICATIONS

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| • Data Analysis School — AI & Machine Learning modules — YÖK / Marmara University, with METU, ITU & Boğaziçi University | 2025 – 2026 |
| • Artificial Intelligence — BTK Academy | 2025 |
| • Machine Learning Applications with Python — BTK Academy | 2025 |
| • Python Programming Language — BTK Academy | 2025 |

LANGUAGES

Turkish: Native · **English**: Professional Working Proficiency · **German**: Basic · **Spanish**: Basic